

UMA302 MATHEMATICS FOR SIGNALS				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To introduce the basic concepts and processing of analog and digital signals.				
Syllabus Introduction: Signals and Systems, Classification of signals, Continuous time signals and its classifications, Standard continuous time signals, Classification of continuous time systems, Discrete time signals and its classifications, Concept of frequency in discrete time signals, Standard discrete time signals, Discrete time systems, Classification of discrete time systems, Nyquist rate, Sampling theorem, Aliasing, Convolution, Correlation. Fourier Transform: Introduction, Condition for existence of Fourier Integral, Fourier Transform and its properties, Energy density and Power Spectral Density, Nyquist Theorem, System Analysis using Fourier Transform. Laplace Transform: Laplace and inverse transform of some common signals, Parseval's theorem, Application of Laplace Transform Properties on analog and digital signals, unilateral Laplace Transform. Analysis and characterization of LTI systems using Laplace transform: Computation of impulse response and transfer function using Laplace transform Z-Transform: Introduction, Region of Convergence (ROC), Properties of z transform. Initial value theorem, Final Value theorem, Partial Sum, Parseval's Theorem, z transform of standard sequences, Inverse z transform, Pole Zero plot, System function of LTI system, Causality and Stability in terms of z transform. Random Signals: Definitions, Random variables, distribution & density functions, mean values & moments, random processes, spectral densities, response of LTI systems to random inputs.				
Laboratory Work (if applicable): Not applicable.				
Course Learning Objectives (CLO) The students will be able to: 1. Enlighten the concepts to analyse signals. 2. Apply Fourier transform for different applications. 3. Apply Laplace transform for different applications. 4. Apply z-transform transform for system characterization. 5. Elucidate the concepts of random signal.				
Text Books 1. Signals and Systems, Oppenheim, A.V. and Willsky, A.S., Prentice Hall of India (1997). 2. Digital Signal Processing: Principles, Algorithms and Applications, Proakis, J.G. and Manolakis, D.G., Prentice Hall (2007).				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.

Reference Books

1. Signal Processing and Linear System, Lathi, B.P., Oxford University Press (2008).
2. Fundamentals of Signals and Systems, Roberts, M.J., McGraw Hill (2007).

Evaluation Scheme	
Evaluation Elements	Weightage %
Mid Semester Test (MST)	25-30
End Semester Examination (ESE)	40-45
Sessional (may include Assignment, Sessional (Includes Regular Lab assessment) and Quizzes Project (Including report, presentation etc.)	30